Semi-Supervised Superpixel-Based Multi-Feature Graph Learning for Hyperspectral Image Data

Madeleine S. Kotzagiannidis and Carola-Bibiane Schönlieb

***Abstract*— Graphs naturally lend themselves to model the complexities of hyperspectral image (HSI) data as well as to serve as semi-supervised classifiers by propagating given labels among nearest neighbors. In this work, we present a novel framework for the classification of HSI data in light of a very limited amount of labeled data, inspired by multi-view graph learning and graph signal processing. Given an *a priori* superpixel-segmented HSI, we seek a robust and efficient graph construction and label propagation method to conduct semi-supervised learning (SSL). Since the graph is paramount to the success of the subsequent classification task, particularly in light of the intrinsic complexity of HSI data, we consider the problem of finding the optimal graph to model such data. Our contribution is two-fold. First, we propose a multi-stage edge-efficient semi-supervised graph learning framework for HSI data, which exploits given label information through pseudo-label features embedded in the graph construction. Second, we examine and enhance the contribution of multiple superpixel features embedded in the graph on the basis of pseudo-label features in an extension of the previous framework, which is less reliant on excessive parameter tuning. Ultimately, we demonstrate the superiority of our approaches in comparison with state-of-the-art methods through extensive numerical experiments.**

***Index Terms*— Graph Laplacian learning (GLL), hyperspectral image (HSI) classification, semi-supervised learning (SSL).**

I. Introduction

THE problem of determining the optimal graph representation of a given dataset has been considered for various tasks in different fields, ranging from signal processing to machine learning, yet remains largely unresolved. Hyperspectral image (HSI) data, with its rich and descriptive spatial and spectral information contained in several hundred bands, encapsulates several layers of dependencies between the high-dimensional pixels and their corresponding labels [1], and a plethora of methods have sought to exploit these under different modeling assumptions.

In the HSI classification, a classifier is sought to assign a class label to each pixel, in light of arising difficulties including high spectral dimensionality, large spatial variability, and limited availability of labels. State-of-the-art approaches comprise kernel methods such as the purely spectral-based support vector machines (SVMs) [2] and spectral–spatial multiple kernel learning [3], which robustly counteract the high spectral dimension. Feature extraction methods, such as IFRF [4], which applies image fusion and recursive filtering, LBP [5], which exploits local binary patterns, and LCMR [6], which uses local covariance matrix representation, have sought to characterize a lower dimensional subspace, which best captures the spectral–spatial information of the data and maximizes discriminability. Furthermore, classical signal processing concepts, such as sparse representation, low-rank, and wavelet analysis, have been notably incorporated into classification methods [1], [7], [8]. More recent methods have gained in complexity by generating multi-stage workflows with different pre- and post-processing levels as well as by combining the strengths of different classifiers in an effort to create more sophisticated models that exploit multiple dependencies of HSI data and counteract the small sample problem through a gradual learning process [1]. In addition, deep learning (DL) models have been employed for HSI feature extraction and classification, including variations of stacked autoencoders (SAEs) [9], convolutional neural networks (CNNs) [10], [11], and generative adversarial networks (GANs) [12].

While the majority of frameworks involve a supervised classifier, due to the time and cost associated with obtaining labeled samples, semi-supervised learning (SSL) has experienced a rapid development by tackling the small sample problem through effective exploitation of both labeled and unlabelled samples [13].

Multiple views or features of data can provide rich insights into the underlying data structure while exploiting diverse information and thus gaining robustness for subsequent tasks. The identification of suitable features, however, represents a problem in itself. Inspired by the field of multi-view clustering [14] and in an effort to incorporate more sophisticated model priors, we adopt the perspective of learning a graph from multiple data features (or views) in order to effectively capture the complexity of hyperspectral data while further integrating the label information into the graph so as to

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exploit prior knowledge of label dependencies. We further resort to superpixel segmentation of the data in order to effectively reduce the complexity of the classification task while simultaneously capturing a first level of spectral–spatial dependencies through the clustering into local homogeneous regions.

In this article, we introduce a novel semi-supervised framework for HSI classification, which involves the design of a graph function that is smooth with respect to both the intrinsic structure of the data, as described via superpixel features, as well as the label space. In particular, by incorporating diverse features that capture spectral-, spatial-, and label-dependencies into the graph building process, we aim to create a rich and comprehensive graph structure, which effectively describes the complex data at hand and therefore facilitates a more accurate classification. The proposed graph learning framework is special due to its edge-efficient analytic solution, known to satisfy graph-optimality constraints, and ability to incorporate multiple features, as well as due to its optimization of both the graph and classification function, resulting in pseudo-label features. In addition, in light of the range of parameters required to tune a multi-feature superpixel graph, we propose a variation of the framework that, instead of incorporating pre-set feature weights, learns them by imposing an additional smoothness functional. The proposed methods are termed multi-feature graph learning (MGL) and parameter-optimal multi-feature graph learning (PMGL). We summarize the main contributions as follows.

1) We extend multi-view graph learning to the domain of superpixels and HSI data, with tunable pseudo-label (feature) generation incorporated into an updateable graph. In particular, we employ an initial soft graph on which labels are first propagated among nearest neighbors to generate pseudo-label features, which are subsequently utilized to inform an improved graph.
2) We propose a pseudo-label-guided framework for HSI feature selection, weighting, and subsequent graph construction, enhanced by dynamic pseudo-label features. To the best of our knowledge, the issue of feature contribution on a multi-feature graph for HSI-data has not yet been tackled, beyond a simple parameter search.
3) We extensively validate our proposed approaches on the basis of three benchmark datasets and demonstrate their superiority with respect to comparable state-of-the-art approaches.

As such, the proposed framework facilitates edge-efficient graph and label learning while flexibly incorporating multiple features that capture spectral-, spatial-, and label-dependencies within the given data. The remainder of this article is organized as follows. Section II discusses related work, Section III explores the preliminaries, and details of the proposed methods are stated in Section IV. Section V presents the experimental results of our methods in comparison with state-of-the-art approaches, and Section VI contains concluding remarks.

II. Related Work

Works in SSL can be categorized into generative models, which employ probabilistic generative mixture models, co-and self-training methods, low-density separation methods, which seek a decision boundary through low-density regions, and graph-based methods [13]. Popular modeling assumptions and concepts have included spectral clustering, data manifolds, and local–global consistency [15], according to which nearby points and points residing on the same cluster or manifold are likely to have the same label [13].

Graph-based SSL methods have become increasingly popular due to the superior modeling capabilities of graphs, particularly as a means to counteract the limited amount of labels available, and have included pixel-based [16] as well as superpixel-based graph constructions [17]–[19], ranging from end-to-end approaches, that utilize graphs for both data modeling and label propagation (LP), to hybrid approaches. We note that many existing methods, such as the edge-preserving filter (EPF) method [20], which makes use of the bilateral filter, exhibit underlying graph-like qualities, while not directly or only in part employing graphs, and as a result have exhibited superior performance. In addition, approaches such as [21] have sought to incorporate label information early on in the workflow, known as pseudo-labeling, which is progressively refined, in an effort to inform data modeling; nevertheless, the graph construction is not necessarily analytic. While identification of the correct model graph is essential for SSL [22], extensive parameter analysis is still inevitable in order to study their influence, as performance is strongly affected by all components of the graph; nevertheless, previous studies have generally not found explorable patterns [22], and stability is preferred over excellence in a narrow parameter range.

A preceding body of work on multi-view clustering has sought to unify the tasks of data similarity learning (i.e., graph construction) and clustering/LP in a joint optimization framework [14], [23], [24], which, to the best of our knowledge, has not been fully investigated for the joint challenge of HSI data and SSL. Graph learning and LP are separate tasks, but recent efforts have opted to combine them, thereby updating and incorporating label information into the graph, under the driving assumption that a single static graph is not sufficient to solve the entire SSL task successfully. Nevertheless, and not least of all for complex data such as HSI, the issue of propagating error noise into each task needs to be addressed. There are multiple ways in which the label information can be incorporated into the graph, which include space fusion approaches, i.e., the graph and label space are fused via the addition of a label correlation matrix [25] or the removal of edges connecting differently labeled nodes [26], and implicit approaches, which instead consider the graph as a function of the distances between label vectors [14].

DL methods have penetrated graph-based classification in the form of graph convolutional networks (GCNs) [27], which usually requires a given pre-constructed graph, such as spectral–spatial GCNs [28], [29], and have been employed to extract features automatically [30]. Under the assumption that the mapping from the feature to the label space is sufficiently smooth, the theoretical relation between LP and GCNs, as instances that propagate labels and features, respectively, has been notably shown to satisfy a smooth-

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ness inequality [31]. Nevertheless, DL- and non-DL methods alike ultimately rely on the optimality of the graph while often performing worse with limited labels. Neural networks have further been employed to learn graphs from scratch (see, e.g., [32]), which inevitably comes at a high computational cost; however, for complex and rich datasets such as HSIs, the formulation of sophisticated (graph) model priors is paramount to the successful performance of any classifier.

Graphs ultimately capture model constraints in the form of linear dependencies between data points (as per the graph Laplacian [33]) and, as such, present versatile modeling tools that can help simplify more complex modeling assumptions. Most graph-based methods operate under the assumption that the given graph is optimal and/or that the given data naturally reside on a known graph, so any subsequent tasks and operations are subject to noise pertaining to an imperfect model. Given a dataset, approaches have opted to either hand-craft the graph or learn it automatically through the minimization of a chosen optimization function. In the former case, this has included local and/or adaptive graph weighting and connectivity schemes, which take into account variations in i.a. (spectral) data density, and are increasingly refined but also bear a risk of distorting neighborhood information [34]. While handcrafted approaches usually lead to higher accuracy, automated ones can be more robust to variable datasets, requiring less parameter tuning while nevertheless being more costly due to lack of an analytic solution. In an effort to combine the advantages of both, we, therefore, seek an analytic solution to a well-defined optimization problem with the possibility of sophisticated parameter learning (i.e., reduction) through regularization.

III. Preliminaries and Prior Work: Graph (Laplacian) Learning

An undirected graph $G = (V, E)$ is characterized by a set of vertices V and a set of edges E, and its connectivity is encapsulated in the symmetric adjacency matrix $\mathbf{W}$, with $W_{i,j} > 0$ if there is an edge between vertices i and j and $W_{i,j} = 0$ otherwise. The non-normalized graph Laplacian matrix is defined as $\mathbf{L} = \mathbf{D} - \mathbf{W}$, where $\mathbf{D} = \text{diag}(\mathbf{W1})$ is the diagonal degree matrix, with $\mathbf{1}$ denoting the vector of 1's. The construction of a graph that optimally represents a given dataset $\mathcal{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$ is a multi-part task, which generally establishes relations of the form $W_{i,j} = A_{i,j}\psi(\mathbf{x}_i, \mathbf{x}_j)$ between data vectors $\mathbf{x}_i, \mathbf{x}_j \in \mathbb{R}^d$, where $\psi : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ is a similarity function and $\mathbf{A}$ is an adjacency matrix, for which $A_{i,j} = 1$ iff $j \in \mathcal{N}_i$ for some pre-determined neighborhood $\mathcal{N}_i$ of node i, and $A_{i,j} = 0$ otherwise. From the smoothness of $\psi(\cdot)$ to the range of influence $\mathcal{N}$, each component is vital in ensuring effective discriminative data representation.

In graph signal processing, graph Laplacian learning (GLL) considers the minimization of the graph Laplacian quadratic form with respect to data (aka graph signal) matrix $\mathbf{X}$

$$\text{Tr}(\mathbf{X}^T\mathbf{L}\mathbf{X}) = \frac{1}{2}\sum_{i,j} W_{i,j}||\mathbf{x}_i - \mathbf{x}_j||_2^2 = \frac{1}{2}||\mathbf{W} \circ \mathbf{Z}||_{1,1}$$

with trace operator $\text{Tr}(\cdot)$ and pairwise distances $Z_{i,j} = ||\mathbf{x}_i - \mathbf{x}_j||_2^2$ and can be alternatively expressed as a weighted sparsity l_1-norm of $\mathbf{W}$, whose minimization enforces connectivity (large $W_{i,j}$) between similar feature vectors $\mathbf{x}_i$ and $\mathbf{x}_j$. Previous works (see, e.g., [35], [36]) have considered variations of the general graph optimization framework

$$\min_{\mathbf{W}\in\mathcal{W}} ||\mathbf{W} \circ \mathbf{Z}||_{1,1} + f(\mathbf{W}) \tag{1}$$

consisting of the graph Laplacian quadratic form and a, possibly sparsity-promoting, regularization term $f(\mathbf{W})$, subject to graph constraints, e.g., $\mathcal{W} = \{\mathbf{W} \in \mathbb{R}_{\geq 0}^{N\times N} : \mathbf{W} = \mathbf{W}^T, \text{diag}(\mathbf{W}) = 0\}$. We select $f(\mathbf{W}) = \alpha||\mathbf{W}||_F^2, \alpha \in \mathbb{R}$, with Frobenius norm $||\mathbf{W}||_F = (\sum_{i,j}|W_{i,j}|^2)^{1/2}$, which controls sparsity by preventing the occurrence of strong edges and, due to the edge-locality of the functional, facilitates the decomposition of the problem into a sum over graph edges and, hence, an analytic solution.

Specifically, the optimization problem of (1) becomes separable for $f(\mathbf{W}) = \alpha||\mathbf{W}||_F^2$ and can be rewritten in row-wise form [14], for row $\mathbf{W}_i$ of $\mathbf{W}$, as

$$\min_{\mathbf{W}_i\mathbf{1}=1, \mathbf{W}_i\geq\mathbf{0}, W_{i,i}=0} \sum_{j=1}^{N} Z_{i,j}W_{i,j} + \alpha\sum_{j=1}^{N} W_{i,j}^2 \tag{2}$$

yielding the closed-form solution

$$W_{i,j} = \left(\eta_i - \frac{Z_{i,j}}{2\alpha}\right)_+$$

with $(x)_+ = \max(0, x)$ and scalar η_i. Since the second term creates a dense edge pattern, one can further enforce k-nearest neighbor (kNN) connectivity by determining the maximal α_i per row such that the optimal $\mathbf{W}_i$ has exactly k non-zeros (see [14] for details). This leads to

$$W_{i,j} = \begin{cases} \dfrac{Z_{i,k+1} - Z_{i,j}}{kZ_{i,k+1} - \sum_{h=1}^{k} Z_{i,h}}, & j \leq k \\ 0, & j > k \end{cases} \tag{3}$$

where entries $\{Z_{i,1}, \ldots, Z_{i,N}\}$ are assumed to be ordered from small to large without loss of generality, and we have $\eta_i = (1/k) + (\sum_{h=1}^{k} Z_{i,h}/2k\alpha_i)$ and $\alpha_i = (k/2)Z_{i,k+1} - (1/2)\sum_{h=1}^{k} Z_{i,h}$. Here, $W_{i,i} = 0$ is enforced and $Z_{i,i} = 0$ appended at the end. Symmetrization is achieved through $\mathbf{L} = \mathbf{D} - (\mathbf{W} + \mathbf{W}^T)/2$.

It has been noted that this approach is computationally efficient due to its analytic solution and in-built sparsity, which does not require oblique tuning of α (instead only requiring the straight-forward number of edges k) and is further scale-invariant with respect to feature vectors $\mathbf{x}_i$.

For different features (views) of type v, we consider $Z_{i,j} = \sum_v c_v Z_{i,j}^v$ with feature coefficients $c_v \in \mathbb{R}$. In order to reduce parameters, it has been proposed to constrain the coefficients to be proportional to pre-set feature-dependent functionals and subject to regularization [23], [24].

IV. Proposed Method: Superpixel-Based Multi-Feature Graph Learning

In the proposed method, several instances of data dependencies are exploited in a multi-level workflow. After conducting an initial spectral dimensionality reduction using PCA [37], we consider *a priori* the segmentation of the HSI into

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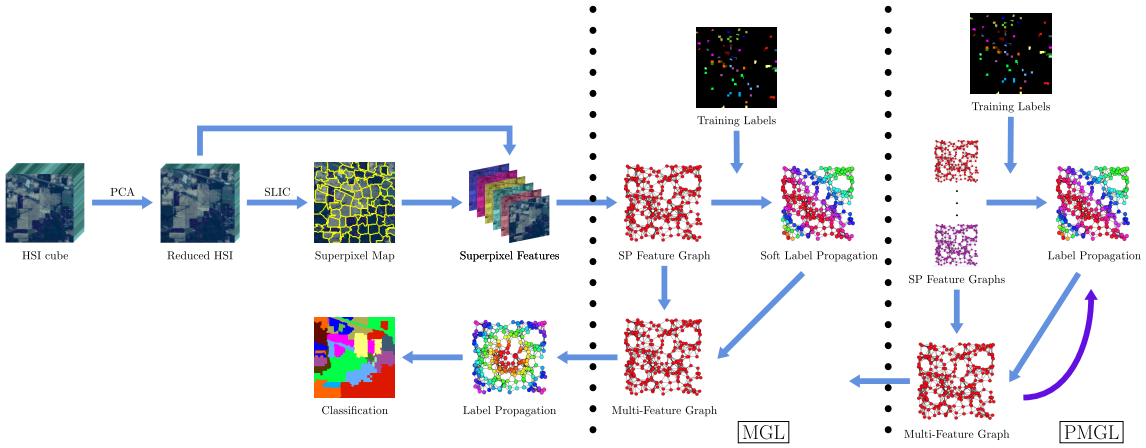


Fig. 1. Main workflow: The proposed MGL and PMGL methods are mainly composed of spectral dimensionality reduction, superpixel segmentation, and feature extraction, followed by respective graph learning stages to create a multi-feature graph, and culminating in the final classification result via LP.

superpixels to define local regions of homogeneous spectral content and to reduce spatial dimensionality. Subsequently, we compute analytic superpixel features, which capture different image properties, and extrapolate given pixel labels to superpixel labels via a simple averaging filter. We then compute an initial graph G_0 and form pseudo-label features through a soft label propagation to nearest neighbors on G_0. This is subsequently updated and refined, before a final graph classifier is applied (see Fig. 1).

A. Superpixel-Based Feature Extraction

Given the raw HSI data cube $\mathbf{I} \in \mathbb{R}^{X\times Y\times B}$, we apply dimensionality reduction in the first instance in the spectral domain using PCA to obtain the reduced $\tilde{\mathbf{I}} \in \mathbb{R}^{X\times Y\times b}$, $b \ll B$. By exploiting the correlations present in the data between neighboring hyperspectral bands, we conduct PCA in order to extract non-correlated features with increased computational efficiency. In particular, PCA is known to keep the most information in its first few components in terms of variance present in the data and is competitive for the purpose of classification due to its lack of parameters and low complexity [38].

Subsequently, the first principal component is used to conduct superpixel segmentation via SLIC [39], resulting in the 2-D superpixel labeling map $\tilde{\mathbf{S}} \in \mathbb{R}^{X\times Y}$ with N superpixels

$$\mathbf{S}_k \text{ s.t. } \mathbf{S}_k = \{\mathbf{S}_{(i,j)} | \mathbf{S}_{(i,j)} = k\}, \quad \tilde{\mathbf{S}} = \cup_{k=1}^{N} \mathbf{S}_k.$$

While it has been established that the goodness and scale of the superpixel segmentation are foundational for the success of subsequent data modeling and classification tasks, it is not the objective of this work to optimize this particular instance of the workflow; as such, we select SLIC [39] as the base superpixel segmentation algorithm and determine the number of superpixels approximately according to [40], which takes into account both the size and resolution of the HSI image (albeit not the scene complexity).

Let $\mathbf{Y} \in \mathbb{R}^{XY\times c}$ denote the initial class indicator matrix with $Y_{i,j} = 1$ if pixel i belongs to class j for c classes. Following superpixel segmentation, we regularize this to $\mathbf{Y}^S \in \mathbb{R}^{N\times c}$ by averaging over the existing pixel labels per superpixel. Specifically, $Y^S_{i,j}$ records the number of pixels per superpixel $\mathbf{S}_i$, which belongs to class j divided by the total number of pixels in $\mathbf{S}_i$.

As statistical descriptors for the superpixels, we consider the features as proposed in [3], comprising the mean feature vector $\mathbf{s}^M_k$

$$\mathbf{s}^M_k = \frac{\sum_{i,j}^{N_k} \tilde{\mathbf{I}}_{(i,j)}}{N_k}, \quad \mathbf{S}_{(i,j)} = k, \ k = 1, \ldots, N$$

which takes a simple average of the N_k pixels per superpixel k, and the spatial-mean feature vector $\mathbf{s}^S_k$

$$\mathbf{s}^S_k = \sum_{i=1}^{J} w_{k,a_i} \mathbf{s}^M_{a_i}, \ w_{k,a_i} = \frac{\exp\left(-\left\|\mathbf{s}^M_{a_i} - \mathbf{s}^M_k\right\|_2^2 / h\right)}{\sum_{i=1}^{J} \exp\left(-\left\|\mathbf{s}^M_{a_i} - \mathbf{s}^M_k\right\|_2^2 / h\right)}$$

which constitutes a weighted sum of the mean feature vectors of adjacent superpixels, given by index set $\mathcal{A}_k = \{a_1, \ldots, a_J\}$ for the k-th superpixel and with pre-set scalar $h \in \mathbb{R}$. Furthermore, we consider the centroidal location of each superpixel as

$$\mathbf{s}^C_k = \frac{\sum_{j=1}^{N_k} l_{k,j}}{N_k}$$

where $l_{k,j}$ denotes the 2-D image coordinate of the k-th superpixel. In particular, $\mathbf{s}^M$ and $\mathbf{s}^S$ reflect the spectral and spatial information, respectively, within and among superpixels, while $\mathbf{s}^C$ is a pure spatial descriptor, which acts as a regularizer in conjunction with the former. These features were selected due to their analytic form and low computational complexity. We note that an optimized graph can only be as good as the extracted features it is built upon, however, the task of feature optimization, in line with superpixel segmentation, represents a problem in itself and is not the focus of this work.

B. Dynamic Graph Learning and LP

In the following, we wish to learn a superpixel HSI graph and conduct classification through LP.

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In an extension of (2), consider the joint optimization problem over graph $\mathbf{W}$ and class indicator matrix $\mathbf{F} \in \mathbb{R}^{N\times c}$

$$\min_{\mathbf{W},\mathbf{F}} \sum_{i,j} \left(Z_{i,j} + \gamma\, Z^F_{i,j}\right) W_{i,j} + \alpha W^2_{i,j}$$
$$\text{s.t.} \quad \sum_j W_{i,j} = 1, \quad W_{i,j} \geq 0, \ \mathbf{F}_l = \mathbf{Y}^S_l \tag{4}$$

with $Z^F_{i,j} = ||\mathbf{f}_i - \mathbf{f}_j||^2_2$, where $\mathbf{f}_i$ denotes the i-th row of $\mathbf{F}$ and $\mathbf{Y}^S_l \in \mathbb{R}^{l\times c}$ is the labeled submatrix of $\mathbf{Y}^S$. Via alternating optimization, the optimal graph $\mathbf{W}$ can be computed according to (3) with $Z_{i,j} \rightarrow Z_{i,j} + \gamma\, Z^F_{i,j}$ and by absorbing α into k, the number of edges per row, while for $\mathbf{F}$, this yields the solution

$$\mathbf{F}_u = -\mathbf{L}^{-1}_{u,u}\mathbf{L}_{u,l}\mathbf{Y}^S_l \tag{5}$$

for the unlabeled superpixel nodes, which is also known as harmonic LP [24], [41]. Here, $\mathbf{L}_{u,u}$ denotes the graph Laplacian submatrix with rows and columns corresponding to unlabeled nodes. The final superpixel labels are assigned via the decision function

$$y_i = \arg\max_j F_{i,j} \quad \forall j = 1, \ldots, c. \tag{6}$$

Accordingly, we require the optimal graph to exhibit smoothness with respect to both the pre-designed superpixel features, as extracted from the HSI data, as well as the label features (rows) of $\mathbf{F}$. The functional $\text{Tr}(\mathbf{F}^T\mathbf{L}\mathbf{F}) = (1/2)\sum_{i,j} Z^F_{i,j} W_{i,j}$, given $\mathbf{L}$, has been employed as a standalone graph classifier (e.g., [15], [41]), while in the present framework, it is further utilized to enrich the graph construction process, rendering it dynamic.

Let $\mathbf{Z}^\upsilon$ with $Z^\upsilon_{i,j} = ||\mathbf{s}^\upsilon_i - \mathbf{s}^\upsilon_j||^2_2$ denote the Euclidean distance matrix between superpixel feature vectors of type (or view) $\upsilon \in \{M, S, C\}$, as previously defined in Section IV-A, and c_υ be the feature weight. In a variation of the above, we propose to employ the superpixel multi-feature dynamic graph G of the basic form as in (3) with weighted pairwise distances $Z_{i,j} = \sum_\upsilon c_\upsilon Z^\upsilon_{i,j}$ and $Z^{\tilde{F}}_{i,j} = ||\tilde{\mathbf{f}}_i - \tilde{\mathbf{f}}_j||^2_2$, the latter of which we define as pseudo-label features

$$\tilde{\mathbf{f}}_i[j] = \sum_k P^0_{i,k} Y^S_{k,j}, \quad j = 1, \ldots, c$$

where $\tilde{\mathbf{f}}_i$ denotes the i-th row of $\tilde{\mathbf{F}} \in \mathbb{R}^{N\times c}$, $\mathbf{W}_0$ is the initial graph based on $Z_{i,j}$, and $\mathbf{P}^0 = \mathbf{D}^{-1}_0\mathbf{W}_0$ is its random-walk normalized version. In particular, $\tilde{\mathbf{F}}$ constitutes one instance of a random walk, as opposed to the fully converged solution in (5). The motivation behind this construction is to provide a soft pre-labeling approach by propagating given labels among their nearest neighbors, as determined through an initial superpixel graph $\mathbf{W}_0$ based solely on HSI superpixel features, thereby merging information from the label and superpixel dependencies. The resulting distance matrix $\mathbf{Z}^{\tilde{F}}$ is then utilized as an additional component to rebuild the graph, with large values penalizing nodes not in the same class, further rendering the graph construction dynamic and implicit. Final label propagation on this graph is conducted via the converged harmonic solution in (5) and class assignment via (6). We summarize the approach in Algorithm 1.

Algorithm 1 Multi-Feature Graph SSL for HSI Data

1: **INPUT:** raw HSI cube $\mathbf{I}$, label matrix $\mathbf{Y}$, parameters: k, $\{c_\upsilon\}_\upsilon$, γ.
2: **OUTPUT:** Classification map.
3: Apply PCA on $\mathbf{I}$ to obtain $\tilde{\mathbf{I}}$.
4: Conduct superpixel segmentation to obtain $\tilde{\mathbf{S}}$ and label regularization to obtain $\mathbf{Y}^S$.
5: Extract superpixel features $\{\mathbf{s}^M, \mathbf{s}^S, \mathbf{s}^C\}$.
6: Compute initial superpixel-feature graph $\mathbf{W}_0$ with $Z_{i,j} = \sum_\upsilon c_\upsilon Z^\upsilon_{i,j}$, pre-set c_υ, $\upsilon \in \{S, M, C\}$, via (3) & symmetrize.
7: Compute pseudo-label features $\tilde{\mathbf{F}} = \mathbf{P}^0\mathbf{Y}^S$ and $\mathbf{Z}^{\tilde{F}}$.
8: Update graph with $\tilde{Z}_{i,j} = Z_{i,j} + \gamma\, Z^{\tilde{F}}_{i,j}$ and symmetrize.
9: Compute unknown labels with graph classifier $\mathbf{F}_u = -\mathbf{L}^{-1}_{u,u}\mathbf{L}_{u,l}\mathbf{Y}^S_l$.
10: Assign final classes via (6).

Remark: The RBF kernel with $W_{i,j} = \exp(-(||\mathbf{x}_i - \mathbf{x}_j||^2_2/\sigma^2))$ constitutes a prominent approach to model graph weights [22] and is incidentally the result of the optimization problem in (1) with $f(\mathbf{W}) = \sigma^2 \sum_{i,j} W_{i,j} \log W_{i,j}$ in normalized form. Nevertheless, performance is strongly affected by the tuning of σ and graph connectivity, the latter of which is not embedded into the graph solution and both of which are nontrivial.

While the proposed approach simplifies the issue of tuning edge connectivity and neighborhood range, thus increasing robustness, it still comprises a range of parameters, which are categorized into k (the number of nearest neighbor edges), $\{c_\upsilon\}_\upsilon$ (superpixel feature weights), and γ (pseudo-label feature weight). As the goodness of the graph is dependent on the goodness of its features, we proceed to simultaneously learn feature weights and update the graph with the goal to help guide as well as minimize uninformed parameter-tuning.

C. Parameter-Optimal Multi-Feature Graph Learning

While a reduction of parameters generally occurs at the sacrifice of performance and cannot replace a thorough parameter search, we investigate the possibility of a pseudo-label feature guided parameter reduction and further propose a variation of the preceding framework, inspired by [23], in an effort to facilitate training and generalizability to diverse datasets.

Consider individual superpixel feature graphs, denoted with adjacency matrix $\mathbf{A}^\upsilon$, whose entries are computed from pairwise feature distances $Z^\upsilon_{i,j}$ using (3) with k edges per row, and initialize the global graph $\mathbf{W} = \sum_{\upsilon=1}^V c_\upsilon \mathbf{A}^\upsilon$ with $c_\upsilon = (1/V)$. We assume that the deviation $r^\upsilon = ||\mathbf{W} - \mathbf{A}^\upsilon||^2_F$ from $\mathbf{W}$ is inversely related to the feature importance c_υ. Following an initial computation of pseudo-label feature $\tilde{\mathbf{F}}$ using (5), which, in contrast to the previous method, utilizes the converged harmonic solution, we update the global graph $\mathbf{W}$ by solving

$$\min_{\mathbf{W}_i\geq 0, \mathbf{W}_i\mathbf{1}=1} \left|\left| \mathbf{W}_i + \frac{(\gamma_1/2)\mathbf{Z}^{\text{WF}}_i - \sum_\upsilon c_\upsilon \mathbf{A}^\upsilon_i}{\sum_\upsilon c_\upsilon} \right|\right|^2_2 \tag{7}$$

per row. Here, $\mathbf{Z}^{\text{WF}} = W \circ \mathbf{Z}^{\tilde{F}}$ denotes the Hadamard product of distance matrix $\mathbf{Z}^{\tilde{F}}$ of $\tilde{\mathbf{F}}$ and a binary mask W,

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which holds the nonzero locations of the current global graph estimate $\mathbf{W}$. As such, when solving (7), the positions of the non-zero graph weights remain fixed, while their values are perturbed, i.e., re-weighted or eliminated according to pseudo-label information. This prevents the formation of noisy edges when $\mathbf{Z}^{\tilde{F}}$ is dense and constitutes an alternative to soft pseudo-labeling (for which we previously employed one random walk). Subsequently, given $\mathbf{W}$ and $\mathbf{A}^{\upsilon}$, we regularize the feature graph weights c_{υ} via an l_2-norm term

$$\min_{\mathbf{c}} \sum_{\upsilon} c_{\upsilon} ||\mathbf{W} - \mathbf{A}^{\upsilon}||_F^2 + \gamma_2 ||\mathbf{c}||_2^2, \text{ s.t. } c_{\upsilon} \geq 0, \quad \mathbf{c}^T \mathbf{1} = 1 \tag{8}$$

which can be simplified to

$$\min_{c_{\upsilon} \geq 0, \mathbf{c}^T \mathbf{1} = 1} \left\| \mathbf{c} + \frac{\mathbf{r}}{2\gamma_2} \right\|_2^2. \tag{9}$$

Notably, (2) can be written in the same form as (7) and (9), which constitutes the Euclidean projection onto the probabilistic simplex [42]; however, as we cannot apply the same kNN simplification, we solve the latter two iteratively, whereby we employ Newton's method to enforce the unity sum constraint. As such, both the graph edge learning and feature weight learning stage are essentially the same.

By recomputing $\mathbf{Z}^{\mathrm{WF}}$ and then $\mathbf{W}$ with corresponding parameter γ_3, we obtain a pseudo-label enhanced graph, which is employed by the graph classifier of (5) to obtain the final solution.

Overall, γ_2 controls the disparity between feature weights while replacing V parameters with one, while γ_1 and γ_3 regulate the pseudo-label contribution at different stages. Here, we employ pseudo-labels in a two-fold way to inform feature contribution as well as to form a separate feature embedded in the graph. While this can be tuned with a single parameter γ_1, in practice, we observe that performance benefits from weighting the steps separately by introducing γ_3, as we will demonstrate in Section V. We summarize the graph learning stage of the approach in Algorithm 2; here, each computed graph is *a posteriori* symmetrized via $(\mathbf{W} + \mathbf{W}^T)/2$. The approach is similar to solving the joint optimization problem of (4) (with respect to $\mathbf{F}$) and (8) alternately, as each step constitutes an optimal closed-form solution; however, we refrain from further iterations between steps **4** and **6** to limit possible noise resulting from pseudo-labeling.

We note that large deviations in (feature) scales and, thus in $\mathbf{r}$, result in binary weights (i.e., single-feature selection), which can be remedied, in part, by tuning γ_2 as well as by refining the feature selection. For this approach, we introduce two composite superpixel feature measures that merge the centroidal feature, which is less informative for a standalone feature graph, with either of the spectral-content features. Specifically, we consider the multiplicative $Z_{i,j}^{\upsilon} \circ Z_{i,j}^{C}$ and additive $Z_{i,j}^{\upsilon} + \lambda Z_{i,j}^{C}, \upsilon \in \{M, S\}$, with υ chosen as per dataset and $\lambda \sim \sigma_{\upsilon}/\sigma_C$, where $\sigma_{\upsilon} = \sum_{i,j} Z_{i,j}^{\upsilon}/N^2$ denotes the scale per feature.

Remark: One could further consider the constraint $r^{\upsilon} = \sum_{i,j} Z_{i,j}^{\upsilon} W_{i,j}$ in (8) as a non-separable way to estimate the feature weights; however, we observe that discrepancies in scaling render the parameter γ_2 more difficult to tune. Instead, we opt to separate the graph into the sum of individual feature graphs. While this bears the bias of reduced global interaction between the features (i.e., in (3), the sum $Z_{i,j} = \sum_{\upsilon} c_{\upsilon} Z_{i,j}^{\upsilon}$ drives the assignment of the nearest k neighbors), we remedy this by incorporating pseudo-labels into the framework as a means to perturb the solution and reinforce inter- and intra-class relations as well as by introducing composite features.

Algorithm 2 Parameter-Optimal Multi-Feature Graph SSL

1: Initialize superpixel-feature graphs $\mathbf{A}^{\upsilon}$ with $\mathbf{Z}^{\upsilon}$ via (3) and $\mathbf{W}^0 = \sum_{\upsilon} c_{\upsilon} \mathbf{A}^{\upsilon}$ with $c_{\upsilon} = \frac{1}{V}$, then symmetrize.
2: Compute pseudo-label features $\tilde{\mathbf{F}}$ via (5) and $\mathbf{Z}^{WF} = W \circ \mathbf{Z}^{\tilde{F}}$.
3: Update graph with $\tilde{Z}_{i,j} = (\frac{\gamma_1}{2} Z_{i,j}^{WF} - \sum_{\upsilon} c_{\upsilon} A_{i,j}^{\upsilon}) / \sum_{\upsilon} c_{\upsilon}$ with pre-set γ_1 by solving (7), symmetrize.
4: Compute feature weights c_{υ} with pre-set γ_2 via (9).
5: Update pseudo-label features $\tilde{\mathbf{F}}$ via (5).
6: Update graph with pre-set pseudo-label weight γ_3 in (7), then symmetrize.
7: Compute final labels via graph classifier $\mathbf{F}_u = -\mathbf{L}_{u,u}^{-1} \mathbf{L}_{u,l} \mathbf{Y}_l^S$ and (6).

V. Experimental Results

A. Dataset Description

We validate our approach on three benchmark HSI datasets.

1) *Indian Pines:* This dataset was gathered by an Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over an agricultural site in Indiana and consists of 145 × 145 pixels with a spatial resolution of 20 m/pixel. The AVIRIS sensor has a wavelength range from 0.4 to 2.5 μm, which is divided into 224 bands, of which 200 are retained for experiments. There are 16 classes, the distribution of which is imbalanced, with Alfalfa, Oats, and Grass/Pasture-mowed containing relatively few labeled samples.
2) *Salinas:* The second dataset was similarly acquired by the AVIRIS sensor over Salinas Valley, California, comprising 512 × 217 pixels with a notably higher spatial resolution of 3.7 m/pixel. Furthermore, 204 bands are retained. The scene contains 16 classes, covering i.a. soils and fields.
3) *University of Pavia:* The third dataset was collected by the Reflective Optics System Imaging Spectrometer (ROSIS), containing 610 × 340 pixels with a spatial resolution of 1.3 m/pixel, the highest of the three datasets. The spectral range from 0.43 to 0.86 μm is divided into 115 spectral bands, of which 103 are retained. The urban site contains 9 classes and covers the Engineering School at the University of Pavia.

B. Experimental Design

In the following, we conduct the experimental evaluation of our methods on the benchmark datasets and demonstrate their superiority compared to state-of-the-art algorithms.

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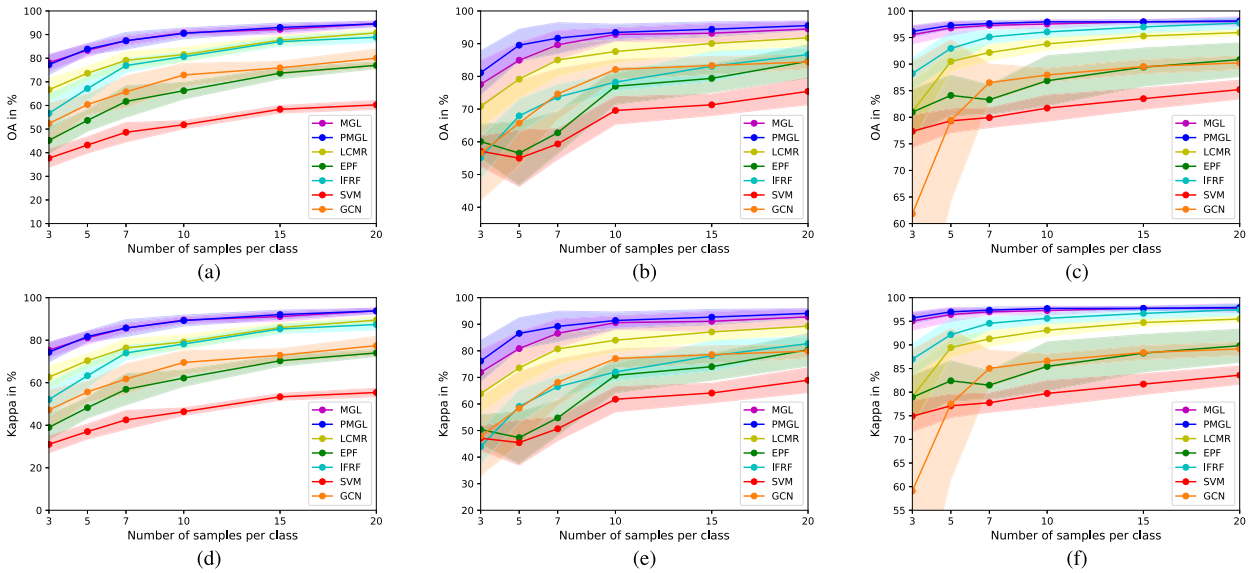


Fig. 2. Comparison of the classification accuracy (OA) and Kappa coefficient of different methods with a varying number of training samples over ten trials. The solid lines represent the mean, while the shaded area covers the standard deviation from the mean. (a) and (d) Indian Pines. (b) and (e) Pavia University. (c) and (f) Salinas.

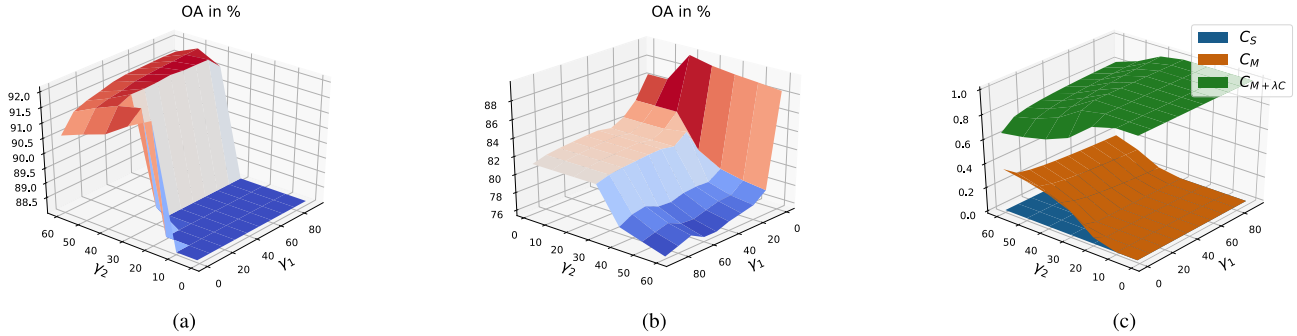


Fig. 3. University of Pavia: results for seven training samples per class, 3-D plot of pseudo-label weight γ_1 versus feature regularization weight γ_2 against OA of PMGL with (a) $\gamma_3 = 0$, (b) $\gamma_3 = 1$, and (c) feature coefficient distribution.

For evaluation, experiments are repeated ten times and performance is assessed on the basis of the average and standard deviation of three quality indices: the overall accuracy (OA), as the percentage of correctly classified pixels, the average accuracy (AA), as the mean of the percentage of correctly classified pixels per class, and the Kappa coefficient (κ), as the percentage of correctly classified pixels corrected by the number of agreements expected purely by chance. Furthermore, we compare the performance with the LCMR [6], the EPF [20], the IFRF [4], and the SVM [2] methods, which, as established state-of-the-art methods, were specifically chosen as comparisons due to their inherent spectral–spatial modeling techniques (with exception of the purely spectral SVM). The SVM algorithm is implemented in the LIBSVM library [43], adopting the Gaussian kernel with fivefold cross validation for the classifier. We further adopt model-parameters as specified in these works. In addition, we conduct comparisons with the GCN [27] method, as a DL-based method. Here, we employ for comparison purposes the same superpixel-segmentation as for our developed methods and consider a kNN superpixel graph ($k = 10$) with a Gaussian kernel, represented by the spectral mean feature at each node. Furthermore, we adopt learning parameters as specified in the original work for two GCN layers with 25 hidden units and choose 10% of given labels for validation to tune hyperparameters and 90% for training the network. The proposed methods are abbreviated as MGL and PMGL.

C. Parameter Specification

For both proposed methods, we conduct PCA on the standardized data to explain 99.8% of the data's variance and employ SLIC [39] for superpixel segmentation with a compactness of 10, where we fix the number K of superpixels per dataset to $K = 1287$ (Indian Pines), $K = 2237$ (Salinas), and $K = 3080$ (University of Pavia). For all graph constructions, we set $k = 10$ as the number of edges per node, and for the spatial mean feature construction, we select $h = 15$.

The following graph-feature/regularization parameters were selected empirically in a coarse to fine search. For the first method (MGL), we employ pseudo-label feature weight $\gamma = 10$ and superpixel feature weights $c_S = 1$, $c_M = 0.5$, and $c_C = 10^{-2}$ (Indian Pines), $c_S = 0.5$, $c_M = 5$, and $c_C = 10^{-5}$ (University of Pavia), and $c_S = 1$, $c_M = 0.1$, and $c_C = 10^{-4}$ (Salinas).

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TABLE I

OA, KAPPA, AA, AND PER CLASS BREAKDOWN IN % WITH SEVEN TRAINING LABELS PER CLASS

Indian Pines							
CLASS	MGL	PMGL	LCMR [6]	EPF [20]	IFRF [4]	SVM [2]	GCN [27]
C1	97.95 ± 1.03	97.95 ± 1.03	$\mathbf{99.23 \pm 1.24}$	95.13 ± 7.69	$\mathbf{99.23 \pm 1.24}$	72.05 ± 13.54	94.1 ± 11.75
C2	66.17 ± 10.17	70.63 ± 11.86	$\mathbf{73.36 \pm 8.24}$	39.32 ± 9.52	65.58 ± 9.98	33.12 ± 4.41	55.22 ± 7.85
C3	76.11 ± 9.73	$\mathbf{86.12 \pm 8.26}$	62.08 ± 8.98	55.65 ± 12.92	76.1 ± 9.16	44.63 ± 6.07	42.25 ± 13.29
C4	$\mathbf{96.17 \pm 4.01}$	92.3 ± 7.54	93.48 ± 6.45	78.13 ± 20.28	87.3 ± 7.99	48.96 ± 12.01	77.09 ± 17.99
C5	$\mathbf{88.84 \pm 9.09}$	88.47 ± 8.83	88.55 ± 8.04	83.8 ± 10.94	80.8 ± 6.47	70.82 ± 10.22	82.1 ± 10.1
C6	$\mathbf{94.54 \pm 10.67}$	$\mathbf{93.43 \pm 8.9}$	86.43 ± 8.34	90.06 ± 8.05	86.02 ± 13.87	68.49 ± 10.61	93.0 ± 4.12
C7	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	98.1 ± 2.46	$\mathbf{100.0 \pm 0.0}$	83.33 ± 15.27	83.81 ± 33.48
C8	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	97.62 ± 4.47	68.3 ± 16.46	99.62 ± 0.71	56.28 ± 9.54	83.74 ± 14.7
C9	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	89.23 ± 12.67	69.23 ± 38.31
C10	$\mathbf{88.05 \pm 7.41}$	$\mathbf{89.97 \pm 7.04}$	76.29 ± 11.93	61.06 ± 17.8	79.19 ± 8.29	45.67 ± 10.66	64.86 ± 11.1
C11	$\mathbf{87.53 \pm 6.94}$	$\mathbf{82.07 \pm 10.0}$	61.55 ± 9.07	48.35 ± 13.16	64.08 ± 12.69	38.08 ± 7.69	44.65 ± 13.73
C12	$\mathbf{86.64 \pm 12.78}$	$\mathbf{93.16 \pm 10.15}$	79.68 ± 7.3	57.59 ± 25.11	68.45 ± 12.47	40.39 ± 11.65	60.85 ± 12.96
C13	$\mathbf{99.49 \pm 0.0}$	$\mathbf{99.49 \pm 0.0}$	99.19 ± 0.9	$\mathbf{99.49 \pm 0.0}$	98.59 ± 1.52	90.86 ± 4.36	98.43 ± 0.69
C14	$\mathbf{97.02 \pm 8.58}$	94.42 ± 8.77	95.86 ± 5.15	76.92 ± 23.07	82.22 ± 7.23	60.51 ± 16.01	85.41 ± 8.25
C15	$\mathbf{94.51 \pm 9.39}$	$\mathbf{94.64 \pm 9.29}$	92.32 ± 9.22	41.42 ± 13.57	86.15 ± 7.41	31.77 ± 7.52	77.73 ± 21.03
C16	93.72 ± 1.74	93.72 ± 1.74	$\mathbf{99.19 \pm 2.57}$	95.23 ± 6.79	97.09 ± 4.04	85.58 ± 10.11	88.6 ± 7.81
OA	$\mathbf{86.75 \pm 2.31}$	$\mathbf{86.93 \pm 2.79}$	77.88 ± 4.07	60.7 ± 6.92	75.77 ± 4.65	47.84 ± 3.29	64.58 ± 5.33
Kappa	$\mathbf{84.96 \pm 2.6}$	$\mathbf{85.19 \pm 3.12}$	75.14 ± 4.51	56.16 ± 7.59	72.84 ± 5.06	42.07 ± 3.6	60.46 ± 5.74
AA	$\mathbf{91.67 \pm 1.31}$	$\mathbf{92.27 \pm 1.11}$	87.8 ± 2.27	74.28 ± 4.98	85.65 ± 2.77	59.99 ± 3.48	75.07 ± 6.48

University of Pavia							
CLASS	MGL	PMGL	LCMR [6]	EPF [20]	IFRF [4]	SVM [2]	GCN [27]
C1	76.23 ± 8.91	78.42 ± 6.35	$\mathbf{79.92 \pm 9.14}$	76.0 ± 11.24	53.47 ± 9.97	62.63 ± 8.75	60.27 ± 5.27
C2	$\mathbf{92.53 \pm 7.9}$	$\mathbf{96.12 \pm 5.36}$	81.38 ± 11.88	59.17 ± 10.17	78.64 ± 6.42	55.55 ± 8.96	75.18 ± 12.13
C3	$\mathbf{87.5 \pm 10.43}$	$\mathbf{90.19 \pm 8.26}$	83.98 ± 10.72	60.88 ± 13.2	55.76 ± 10.56	55.05 ± 9.57	64.4 ± 8.86
C4	84.03 ± 7.83	84.76 ± 6.85	$\mathbf{93.72 \pm 4.43}$	86.49 ± 5.0	65.45 ± 16.27	88.46 ± 5.49	74.87 ± 1.33
C5	93.27 ± 6.35	93.68 ± 5.89	97.2 ± 7.54	96.76 ± 10.23	$\mathbf{99.13 \pm 0.62}$	96.63 ± 8.49	88.77 ± 0.93
C6	$\mathbf{94.94 \pm 6.51}$	$\mathbf{98.23 \pm 2.63}$	84.82 ± 8.41	63.46 ± 13.17	83.4 ± 7.23	57.34 ± 9.54	87.04 ± 1.76
C7	$\mathbf{96.02 \pm 1.84}$	$\mathbf{99.46 \pm 0.26}$	85.15 ± 12.58	94.32 ± 9.97	77.41 ± 14.24	86.22 ± 8.86	93.69 ± 1.42
C8	$\mathbf{82.41 \pm 19.95}$	$\mathbf{90.71 \pm 5.06}$	71.03 ± 9.2	74.57 ± 15.13	71.8 ± 10.57	64.32 ± 10.18	78.81 ± 4.83
C9	92.96 ± 3.37	95.9 ± 2.03	94.46 ± 2.06	$\mathbf{99.78 \pm 0.56}$	50.54 ± 10.7	99.66 ± 0.58	93.36 ± 2.06
OA	$\mathbf{88.7 \pm 3.56}$	$\mathbf{92.08 \pm 2.48}$	82.58 ± 5.74	68.81 ± 4.31	72.63 ± 3.09	63.15 ± 3.64	75.42 ± 5.55
Kappa	$\mathbf{85.3 \pm 4.44}$	$\mathbf{89.61 \pm 3.11}$	77.77 ± 6.74	61.27 ± 4.75	64.86 ± 3.7	54.59 ± 3.77	68.85 ± 6.07
AA	$\mathbf{88.87 \pm 2.35}$	$\mathbf{91.94 \pm 1.6}$	85.74 ± 2.64	79.05 ± 4.05	70.62 ± 2.78	73.98 ± 2.75	79.6 ± 1.72

Salinas							
CLASS	MGL	PMGL	LCMR [6]	EPF [20]	IFRF [4]	SVM [2]	GCN [27]
C1	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	99.89 ± 0.21	99.77 ± 0.5	98.23 ± 5.61	97.85 ± 1.26	$\mathbf{100.0 \pm 0.0}$
C2	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	89.79 ± 5.67	99.74 ± 0.4	96.36 ± 4.47	98.14 ± 1.42	99.63 ± 0.19
C3	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	98.55 ± 2.5	86.26 ± 19.94	99.97 ± 0.08	81.24 ± 18.02	99.26 ± 0.97
C4	94.18 ± 5.92	94.17 ± 8.91	$\mathbf{100.0 \pm 0.0}$	99.88 ± 0.13	98.61 ± 2.63	99.15 ± 0.59	90.84 ± 6.76
C5	93.28 ± 4.31	94.72 ± 0.0	96.53 ± 0.69	$\mathbf{97.62 \pm 1.81}$	92.43 ± 5.1	96.07 ± 2.37	90.64 ± 2.43
C6	$\mathbf{99.59 \pm 0.07}$	$\mathbf{99.59 \pm 0.06}$	99.02 ± 0.99	99.51 ± 0.79	99.53 ± 0.65	98.02 ± 2.1	99.34 ± 1.28
C7	$\mathbf{99.89 \pm 0.06}$	$\mathbf{100.0 \pm 0.0}$	97.9 ± 2.15	99.79 ± 0.07	98.82 ± 3.33	98.57 ± 0.93	93.96 ± 7.09
C8	$\mathbf{98.45 \pm 0.71}$	$\mathbf{97.46 \pm 1.84}$	85.58 ± 5.54	64.96 ± 18.35	85.93 ± 7.01	57.35 ± 11.09	69.01 ± 16.0
C9	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	92.23 ± 9.51	99.49 ± 0.61	99.94 ± 0.15	98.23 ± 0.91	99.61 ± 0.61
C10	89.47 ± 7.41	90.37 ± 5.68	$\mathbf{97.44 \pm 1.37}$	89.24 ± 9.66	97.16 ± 3.84	80.48 ± 9.14	85.06 ± 10.39
C11	95.57 ± 5.16	97.67 ± 0.85	$\mathbf{99.91 \pm 0.08}$	97.15 ± 2.03	96.03 ± 3.32	87.51 ± 3.7	79.63 ± 9.88
C12	97.59 ± 0.31	97.59 ± 0.31	99.23 ± 2.38	$\mathbf{100.0 \pm 0.0}$	98.3 ± 1.45	96.18 ± 3.08	96.53 ± 0.74
C13	97.67 ± 0.62	97.48 ± 0.36	98.26 ± 0.85	$\mathbf{98.92 \pm 0.38}$	95.73 ± 5.84	98.22 ± 0.6	98.4 ± 0.62
C14	92.91 ± 4.8	94.65 ± 2.95	93.13 ± 5.84	95.3 ± 1.66	$\mathbf{96.32 \pm 3.28}$	88.86 ± 2.69	94.01 ± 5.27
C15	$\mathbf{96.32 \pm 1.28}$	$\mathbf{98.23 \pm 0.97}$	82.71 ± 7.54	71.74 ± 21.97	93.98 ± 3.76	59.05 ± 12.78	77.42 ± 11.99
C16	$\mathbf{100.0 \pm 0.0}$	$\mathbf{100.0 \pm 0.0}$	91.23 ± 7.49	94.71 ± 4.17	98.12 ± 3.02	92.33 ± 4.68	95.55 ± 3.19
OA	$\mathbf{97.67 \pm 0.48}$	$\mathbf{97.93 \pm 0.45}$	91.99 ± 1.34	87.14 ± 3.22	94.89 ± 1.7	81.98 ± 1.54	87.54 ± 2.02
Kappa	$\mathbf{97.41 \pm 0.54}$	$\mathbf{97.7 \pm 0.5}$	91.1 ± 1.49	85.73 ± 3.57	94.32 ± 1.89	80.01 ± 1.69	86.17 ± 2.19
AA	$\mathbf{97.18 \pm 0.69}$	$\mathbf{97.62 \pm 0.65}$	95.09 ± 1.01	93.38 ± 2.24	96.59 ± 0.85	89.2 ± 1.5	91.81 ± 1.05

For the second method (PMGL), we employ $\gamma_1 = 0$, $\gamma_2 = 30$, $\gamma_3 = 1$ (Indian Pines), $\gamma_1 = 20$, $\gamma_2 = 40$, and $\gamma_3 = 0$ (University of Pavia) and $\gamma_1 = 1$, $\gamma_2 = 30$, and $\gamma_3 = 1$ (Salinas). Furthermore, for the latter method, we construct three individual feature graphs, respectively, based on the following feature distances: $\{\mathbf{Z}^M, \mathbf{Z}^S, \mathbf{Z}^C \circ \mathbf{Z}^S\}$ (Indian Pines), $\{\mathbf{Z}^M, \mathbf{Z}^S, \mathbf{Z}^M + \lambda \mathbf{Z}^C\}$ (University of Pavia), and $\{\mathbf{Z}^S, \mathbf{Z}^M + \lambda \mathbf{Z}^C, \mathbf{Z}^S + \lambda \mathbf{Z}^C\}$ (Salinas), which were deemed to summarize most effectively the main properties of the different HSI scenes.

D. Experimental Results and Discussion

Performance is evaluated using a very limited amount of labels for training, with rates of $3-20$ samples per class, which are randomly selected, in several stages of experiments, abbreviated as **E1**–**E5**. In the first instance, we conduct numerical evaluations and comparisons of classification accuracy of the proposed methods with state-of-the-art approaches, as detailed above, followed by an evaluation of the corresponding visual classification maps.

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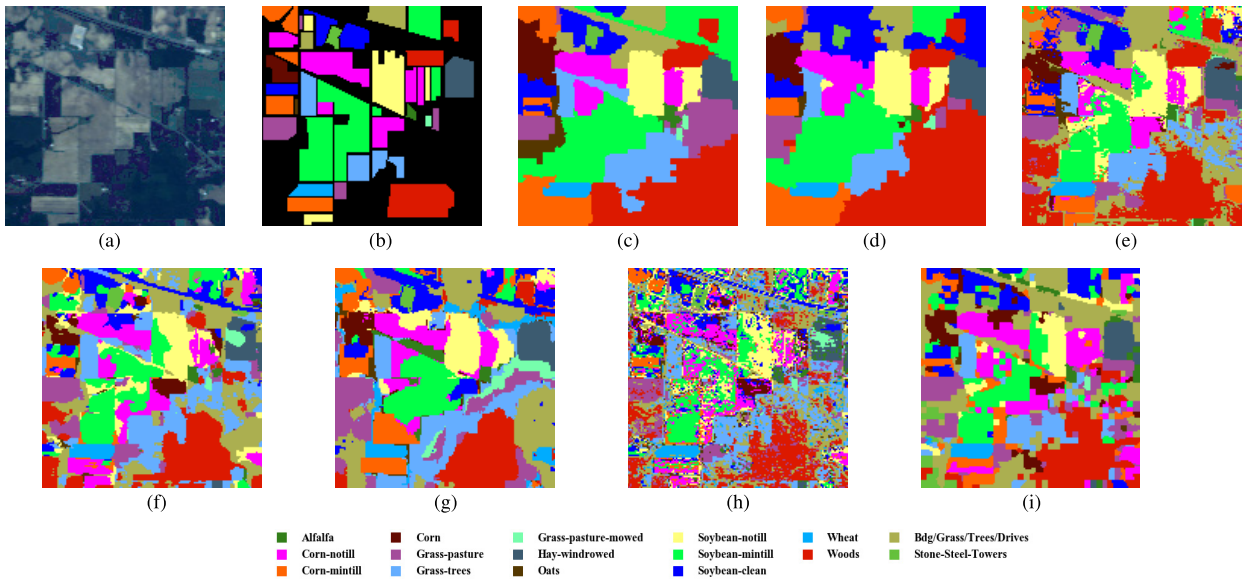


Fig. 4. Indian Pines: (a) color composite, (b) ground truth, and (c)–(i) classification maps produced using seven labeled samples per class: MGL, PMGL, LCMR, EPF, IFRF, SVM, and GCN, respectively.

Finally, we compare the computational times of the different methods.

1) E1: We begin by evaluating the OA and Kappa coefficient in comparison using a reduced label rate of 3–20 randomly selected labels per class, whose results are graphically shown in Fig. 2 for the three benchmark datasets. We observe that both proposed methods consistently outperform the other methods over the entire range of label rates for all three benchmark datasets. In particular, the performance gain is highest for lower label rates, signifying that the proposed multi-feature graph-based methods perform strongly even when extremely few labels are available as a result of their superior model. LCMR and IFRF form the closest competitors for the Indian Pines and Salinas datasets with a maximum gap of approximately 10% and 8%, respectively, to the closest competitor at three labeled samples, with LCMR being the dominant competitor for the more complex Pavia University dataset with a gap of 10% to PMGL. Furthermore, among the two proposed methods, in the lower label limit of the Pavia dataset, PMGL exhibits up to 4% gain in OA performance, while for the other datasets, this gain is vanishingly small with the two methods performing comparably. This indicates that a more refined parameter selection can be beneficial for structurally complex datasets. The GCN-based method on the other hand fails to come close to performances on the upper level and particularly struggles in the lower label limit, exemplifying the challenges of the training process and downside of a less refined graph.

2) E2: To demonstrate the influence of the pseudo-labels and feature coefficients in PMGL on performance, we consider the OA at a fixed label rate of seven samples per class with varying feature parameters over ten trials. In particular, in order to illustrate the interaction between pseudo-label contribution and feature contribution, we fix pseudo-label parameter γ_3 while varying γ_1, along with the feature weight-regularization parameter γ_2, and consider the mean OA in a 3-D plot. We report the results for the University of Pavia dataset, as the most structurally complex of the three, in Fig. 3 with the OA plotted against γ_1 and γ_2 and $\gamma_3 = 0$ in Fig. 3(a), and $\gamma_3 = 1$ in Fig. 3(b), with Fig. 3(c) showing the resulting feature coefficient distribution for Fig. 3(a). We observe that in Fig. 3(a), OA is highest when both γ_1 and γ_2 are increased, generating a perturbation toward more evenly distributed coefficients, which, as shown in Fig. 3(c), corresponds to the gradual matching of composite coefficient $c_{M+\lambda C}$ and spectral mean coefficient c_M, while the contribution of the spatial mean coefficient c_S is negligible. When additionally, the pseudo-label feature is incorporated into the graph via γ_3, the coefficient distribution for the best OA changes, instead overall moving toward one dominant superpixel feature. While we observe interactions between pseudo-label and superpixel features that drive performance, the feature regularization parameter is ultimately dependent on the dataset and the selected features at hand.

3) E3: For each dataset, we use seven labeled samples per class and run the methods again to calculate the OA, Kappa coefficient, AA, and a full class by class accuracy breakdown over ten trials, as shown in Table I. For all three datasets, the proposed methods MGL and PMGL consistently outperform the competitors in OA, Kappa coefficient, and AA and for the majority of per class accuracies. For the Indian Pines dataset, PMGL performs only slightly better than MGL in the first three measures and around 10% improvement in OA over its closest competitor, LCMR. In the case of the University of Pavia dataset, the gain of PMGL over MGL is even larger with 3%–4%, followed by LCMR with 10% improvement in OA. Finally, for the Salinas dataset, MGL and PMGL perform comparatively well, with a gain of 3% over their closest competitor, IFRF. Overall, we observe that while PMGL utilizes more intricate relations and selective feature contributions, MGL is still close in performance, with a gain of the former becoming more evident for increasingly complex datasets, such as Pavia University.

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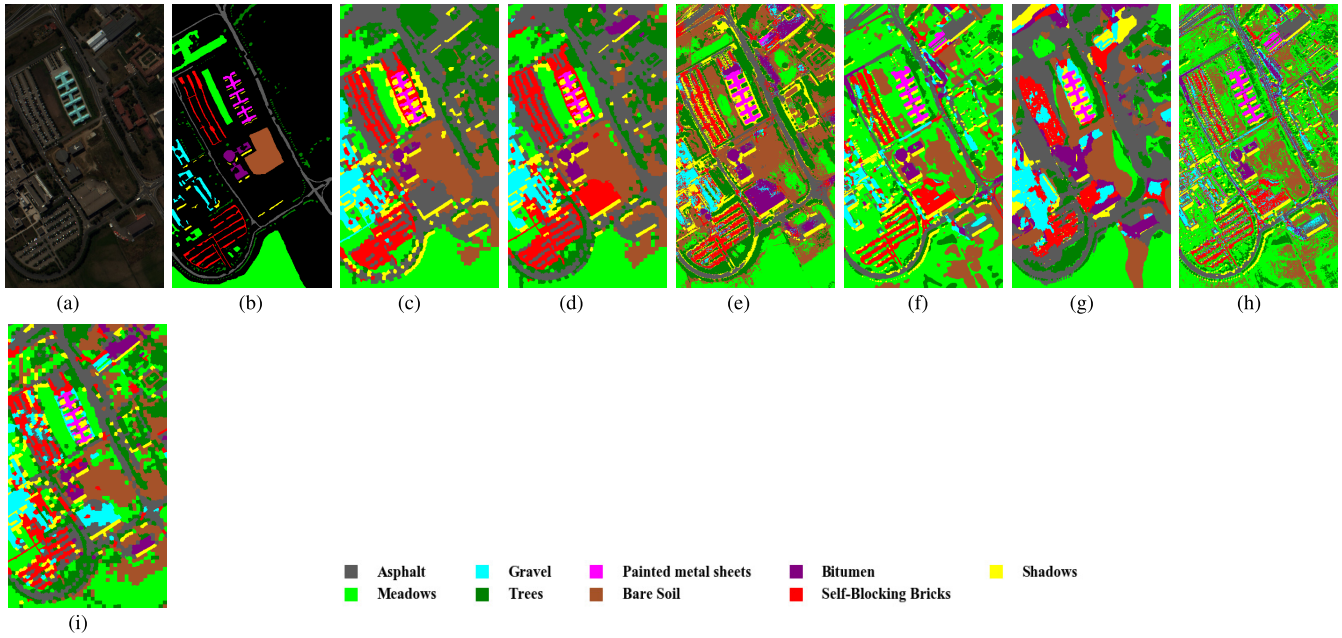


Fig. 5. Pavia University: (a) color composite, (b) ground truth, and (c)–(i) classification maps produced using seven labeled samples per class: MGL, PMGL, LCMR, EPF, IFRF, SVM, and GCN, respectively.

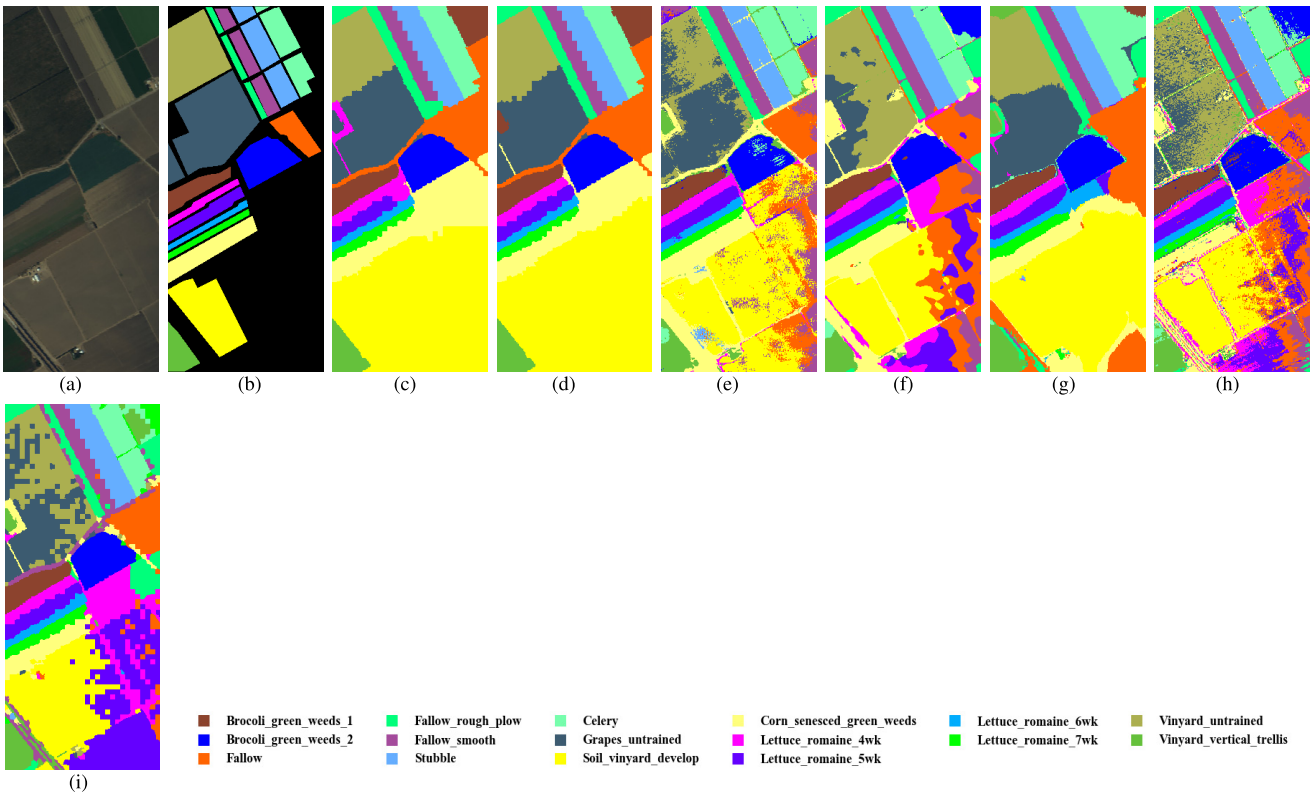


Fig. 6. Salinas: (a) color composite, (b) ground truth, and (c)–(i) classification maps produced using seven labeled samples per class: MGL, PMGL, LCMR, EPF, IFRF, SVM, and GCN, respectively.

4) E4: Furthermore, we show the full classification maps produced for training with seven samples per class for all methods in comparison. In Fig. 4, classification maps for the Indian Pines dataset illustrate increased smoothness and local homogeneity for the proposed graph-based MGL and PMGL, exemplifying their superiority. Their closest competitor, LCMR, exhibits more noisy regions. Fig. 5 shows the University of Pavia classification maps, which as the most structurally complex scene of the three with some scattered classes, similarly exhibits smooth yet spatially refined classification results for the proposed MGL and PMGL, despite the inherent crudeness of superpixel segmentation and label regularization. Finally, for the Salinas dataset in Fig. 6, which due to its locally homogeneous regions and overall spatial regularity represents a simpler scene, the proposed methods still manage

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TABLE II
COMPUTATIONAL TIME (IN SECONDS) WITH SEVEN TRAINING LABELS PER CLASS

Dataset	MGL	PMGL	LCMR [6]	EPF [20]	IFRF [4]	SVM [2]	GCN [27]
Indian Pines	1.71	2.34	9.81	4.06	1.33	4.10	3.13
University of Pavia	8.39	12.08	97.14	3.73	3.33	3.44	18.35
Salinas	5.54	7.47	53.20	7.31	2.56	7.04	11.49

to improve over existing methods, achieving near-perfect classification.

It becomes evident that the use of superpixels, ensuring local homogeneity, as well as that of graphs, for refined local and global modeling, which incorporates pixel- and superpixel-level as well as spectral–spatial-label dependencies, facilitates a significant performance gain. Notably, EPF and IFRF employ a spectral–spatial filtering approach, which can be likened to graph filtering, with the distinction that the latter is more flexible and versatile to model; nevertheless, the use of the SVM classifier in some competitors, as opposed to a graph-filter in the latter case, contributes to a comparatively decreased accuracy due to the noise of the spectral-based SVM.

5) E5: At last, we consider the average computational cost (measured in seconds) for each method, evaluated for experiment **E3** at a label rate of seven samples per class, which is shown in Table II. Here, we ran all tests under the same conditions using an Intel Core i7-9750H processor and 16 GB of RAM. Even though IFRF requires the least computation time for all datasets (followed by SVM and EPF for the Pavia dataset), we observe a relatively low computational cost for the two proposed methods, as a result of the spatial reduction into superpixels and analytic graph design, with MGL being competitive on the Indian Pines and Salinas datasets and PMGL being marginally slower due to its repeated LP computations. Nevertheless, in light of their far superior classification performance, both MGL and PMGL prevail as both accurate and relatively efficient for the considered experiments.

VI. CONCLUSION

In this work, we have developed a pseudo-label-guided and superpixel-based graph learning framework for semi-supervised classification of HSI data. Specifically, we have presented two methods: while the former constructs a single dynamic graph by fusing different superpixel features along with a pseudo-label feature, the latter constructs a global graph as the sum of individual feature graphs whose contribution is informed through pseudo-label regularization. We have demonstrated on the basis of benchmark datasets the superiority in performance through quantitative and qualitative results in comparison to state-of-the-art methods, particularly in the small labeled sample limit.

The incorporation of multiple features as well as label information through pseudo-label features into the graph facilitates refined modeling of the complex dependencies present in HSI data and ultimately leverages these for an improved classification performance. Furthermore, the multi-stage workflow, which employs superpixels and a flexible, inherently sparse graph design with the option to reduce parameters through regularization, is versatile by allowing for multiple components and exploiting multiple levels of spectral–spatial and label dependencies. Nevertheless, it remains a challenge to completely eliminate parameters while maintaining competitive performance as well as to select the ideal features upon which the goodness of subsequent steps depends; ultimately, the process of feature selection/extraction and feature weighting presented is not exhaustive. In our future work, we wish to explore automated DL approaches for both feature selection and graph construction, which can be guided by sophisticated model-based priors.

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